Response to the internal review

We thank the reviewer for a careful reading. We ran 86 further training runs to answer the major points: an IID control that deals training windows rather than homes to clients, runs of up to 100 rounds, a load-shape grouping, a client-only baseline, calendar inputs, a proximal grid that includes $\mu=0$, two more seeds for the heterogeneity, participation and precision settings, and two further cohorts. The title is softened to "Fragmentation outweighs heterogeneity in the accuracy cost of federated household load forecasting". The 85 % traffic claim is withdrawn and the adaptive proximal rule is moved to the supplement. Every number below is generated from the result files, as in the manuscript.

No.	Comment	Response	Where
1	Overall: headline claim not proven; fragmentation, heterogeneity and budget are entangled. Add the controls or soften the title.	Both. The IID control (windows dealt at random) kept 78–83 % of the gap at $K\geq20$ with clients that do not differ, so fragmentation is the larger cost; the regression now finds a significant but smaller heterogeneity term, and longer training narrows the gap. The title was softened from "not heterogeneity" to "outweighs heterogeneity", and the abstract states all three effects.	Title; Abstract; Sections 4.2-4.3
2	Minor: school-level affiliation only.	Done: School of Computer Science and Engineering, Vellore Institute of Technology.	Title page
3	Minor: remove the personal Gmail address.	Removed; only institutional addresses remain.	Title page
4	Check: FedAvg with one client is centralised by construction; present as a sanity check.	Now described as a sanity check of the implementation, labelled so in Table 1, and removed from the contributions.	Section 4.1; Table 1
5	Check: the traffic result holds for FedAvg only and rests on a first-crossing count.	The 85 % claim is withdrawn. With five seeds the increase at $\varepsilon=10$ % is 48 %, and the IID control needs more traffic than random grouping (17.4 against 12.7 MB), so the difference is within seed variation. The abstract no longer makes a traffic claim for heterogeneity.	Abstract; Section 4.3
6	Check: state the configuration or range of the 66 % to 48 % claim.	The abstract now states $K=20$, $\alpha=0.3$ and uses the 1 % margin (55 % to 39 %); Section 4.7 gives the range over groupings and $K=100$.	Abstract; Section 4.7; Table 4
7	Major 6 (1): centralised beats persistence by only 8.9 %; add calendar inputs or a stronger model; report MASE.	Calendar inputs were added to centralised, local and FedAvg ($K=20$, 100) arms. They did not help (centralised 0.0811 to 0.0815 kWh; FedAvg $K=100$ worse). MASE is reported: centralised 0.987, persistence 1.074. We state in the limitations that single-home forecasts at 30 minutes are close to persistence, so absolute gains are small.	Section 4.4; Table S8
8	Minor: notation t and k used twice.	t is now the half-hour, r the round, k the client and G the number of k-means clusters.	Section 3
9	Major 6 (2): one cohort;	Two further cohorts of 100 homes (selection	Section 4.4;

	add cohorts.	seeds 43 and 44, sharing three or four homes with the main cohort). The gap grew with K in all three; at K=100 it ranged from 3.8 to 10.7 %. Local-only beat centralised in one cohort, which we report.	Table S5
10	Minor: local-only budget.	At most 30 epochs with early stopping on validation MAE, patience 4.	Section 3.3
11	Minor: how Dirichlet and cluster groupings map homes; equal sizes?	Each grouping is now described step by step; all clients hold exactly 100/K homes.	Section 3.3
12	Major 3: heterogeneity range narrow; composite H dominated by mean load; justify H, one index.	The composite is dropped (its level term made up 94 % of it). HJS (base 2, so bounded by 1, not ln 2) is used throughout, with HCV reported separately. A shape-sorted grouping was added (HJS=0.0213, the widest split available, with unchanged error 0.0857 kWh); the reachable range remains small, and the discussion limits the conclusion accordingly.	Section 3.3; Table 3; Section 5
13	Minor: state the raw-data encoding; show 2-4 bytes and CSV.	12 bytes per reading is stated, and the comparison is given at 2 bytes (2.3 MB), 4 bytes (4.7 MB), 12 bytes (14.0 MB) and CSV (58.1 MB). Thirty rounds at K=20 still move more (82.6 MB).	Sections 3.3 and 4.6
14	Major 1: gap may be under-training; plot per-round validation MAE per K, report chosen round, run K >= 20 to a plateau, state whether Adam is reset.	Adam is reset at every local update (stated). Steps per round and the chosen round are tabulated, and the curves plotted (Table S2, Fig. S1). Runs of up to 100 rounds with patience 25 narrowed the gap to 2.7, 5.9 and 7.4 % at K=20, 50, 100. K=50 plateaued; at K=20 and 100 the best round was still near the end, which we state.	Sections 3.1, 3.4, 4.2; Fig. 2; Table S2; Fig. S1
15	Minor: what the seeds vary.	Initialisation, grouping or Dirichlet draw, client sampling and stragglers, and mini-batch order; the cohort is fixed.	Section 3.4
16	Check: WAPE never reported; MAPE* unexplained.	WAPE is reported for every arm in Table 1; MAPE is dropped from the paper and from Fig. 1, which now lists MAE, RMSE, WAPE and MASE.	Table 1; Fig. 1
17	Minor: seeds in the per-household tests; Holm family.	Household MAE is averaged over seeds before testing; Holm is applied over all rows of Table S1.	Section 3.4; Table S1
18	Check: 'every LSTM beat both naive references' false for K = 100 MAE.	Corrected: FedAvg with one home per client is worse than persistence in MAE (0.0894 against 0.0890 kWh).	Section 4.1
19	Minor: p-values lack '='; hyphens instead of minus signs.	All p-values now read '= value' or '< 10^-4', and tables use the minus sign.	Throughout
20	Major 2: fragmentation and heterogeneity confounded; add IID control; H_JS for every K, log base, CIs.	The window control was added for K=5–100 (Table 2, Fig. 2), with HJS for every K and both partitions. The regression now uses n=54 runs and reports 95 % CIs: ln K 1.37×10^{-3} (1.18 to 1.55), HJS 54.9×10^{-3} (38.1 to 71.7). Both are significant; over the tested ranges the K effect is	Sections 3.3, 4.2; Table 2; Fig. 2

		about four times the heterogeneity effect (0.0041 against 0.0011 kWh). Base-2 logarithms are stated.	
21	Minor: remove titles inside Figs. 2-4.	Removed; panels carry only (a), (b).	Figures
22	Major 4: B@eps looks like a threshold artefact; show curves, vary eps, add seeds and intervals; FedAvg only.	Curves are shown (Fig. 3a), $\varepsilon \in \{5,10,15,20\}$ % with per-seed points (Fig. 3b), five seeds, 95 % intervals, the number of seeds that reached each threshold, and a threshold-free excess-error measure. The claim is withdrawn as above and limited to FedAvg.	Sections 3.3, 4.3; Table 3; Fig. 3
23	Major 7: boundary optimum for mu; widen the grid to 0 or move HA-FedProx out of the contributions.	Both. The grid now includes $\mu=0$ and 10^{-4}; $\mu=0$ (FedAvg) is best (0.07238 against 0.07259 kWh for the best positive μ). HA-FedProx is removed from the contributions and reported in Supplementary S3.	Section 4.5; Tables S3-S4
24	Minor: give n for the Spearman test.	n=30 runs is given (and n for the ρ test in S3).	Section 4.3
25	Minor: why are 16-bit updates and partial participation better?	Rechecked with five seeds; both are slightly better (16-bit for 92 % of households, median -0.0001 kWh). We describe this as a plausible mild regularisation effect below 1 % of MAE, not a gain to exploit.	Section 4.6
26	Check: treat non-significant results consistently.	All results with $p_{Holm} \geq 0.05$ are now described as not significant, without implying equivalence (stated in Section 3.4).	Sections 3.4, 4
27	Check: add results by consumption quintile.	Added (Table S6); the text gives the top-quintile medians.	Section 4.7; Table S6
28	Minor: use a margin for 'harmed'.	A 1 % margin is used, with the share helped by more than 1 % (Table 4).	Section 4.7; Table 4
29	Minor: zoom the x-axis of the benefit CDF.	Zoomed to the 2nd-98th percentile range; the figure moved to the supplement to meet the page limit.	Fig. S2
30	Minor: cite the local-SGD literature and test the mechanism with the controls.	Stich (2019), Woodworth et al. (2020, two papers) and Reddi et al. (2021) are cited; the IID control and the budget runs test the mechanism.	Sections 2.3, 5
31	Check: Briggs et al. Table II had centralised worse than localised; discuss.	Confirmed and discussed: their models (two 20-unit layers, hourly sequences of 6-24 steps, up to 500 epochs) and per-sequence averaging differ. We also report that local-only beat centralised in one of our new cohorts.	Sections 2.1, 5
32	Check: RMSE claim not supported at K = 50 and 100.	The RMSE statement was removed.	Section 5
33	Minor: Taïk's '1.9 kb' looks like a unit slip.	Stated in the text with the weight count (about 4.8 x 10^5 weights, 1.9 MB); the template has no footnotes, so it is a sentence.	Section 5
34	Major 5: pooled clients weaken the privacy premise; add a client-only baseline; who is trusted.	Added for K=5 and K=20 (each client trains alone, with early stopping, on the homes it pools). Plain FedAvg was not significantly better than client-only training (better for 54 and	Sections 3.3, 4.4, 5

		47 % of households, p =0.073 and =0.392); with local fine-tuning it was (66 and 63 %, p <10^{-4} and =0.013). This is now in the abstract and contributions. The discussion states that pooling needs a trusted holder (building gateway, energy community, supplier) and that only one home per client applies otherwise.	
35	Minor: insert the repository URL.	Inserted (https://github.com/Pushkal-Gupta/Short-Term-Household-Energy-Load-Forecasting); the repository will be made public before submission.	Data availability